Discrete Electronics Project: Class AB Audio Amplifier Design and Simulation Report

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***Abstract*—This paper presents the design and simulation of a discrete-component Class AB audio amplifier developed by Analog One. The system combines signal conditioning, voltage amplification, and a power stage to drive an $8\,\Omega$ loudspeaker. The design uses commercial component values and considers signal quality, transistor bias, and heat dissipation. The main calculations establish a $20\,\mathrm{W}$ output reference. In the recorded reference test, a $20\,\mathrm{V}$ peak input at $10\,\mathrm{kHz}$ produced $22.8\,\mathrm{W}$ of average load power. Calculated requirements and simulation readings are presented in separate tables to support a clear assessment of the design.**

***Index Terms*—Class AB, audio amplifier, discrete circuits, MOSFET, circuit simulation.**

I. Project Description

The project develops an audio amplifier that increases the level of an incoming audio signal and supplies the power needed by an $8\,\Omega$ loudspeaker. Developed by Analog One, the design aims to reproduce the signal at a useful output level while limiting distortion and power losses.

The amplifier contains three stages. The preamplifier conditions the incoming signal, the voltage amplifier increases its amplitude, and the power stage provides the current required by the loudspeaker. The Class AB output arrangement uses two complementary transistors that share the audio waveform. A small resting current helps smooth the transition between the positive and negative portions of the signal.

The design combines calculations, commercial component values, and circuit simulation. The evaluation considers output power, signal transfer, bias adjustment, and heat dissipation, together with the interaction between the stages.

II. Block Diagrams

A. Overall Amplifier

Figure 1 shows how the signal moves through the amplifier. Each stage has a specific role, from preparing the incoming signal to delivering power to the loudspeaker.

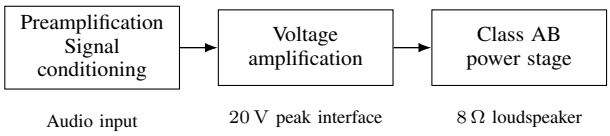


Figure 1. Main stages of the audio amplifier.

B. Preamplification Stage

C. Voltage Amplification Stage

D. Class AB Power Stage

The Class AB arrangement was selected because it combines a smooth transition between the two portions of an audio signal with moderate resting power consumption. Its overlapping conduction helps reduce crossover distortion, which is particularly relevant when reproducing small changes in the waveform [1, sec. 7–2].

The selected circuit also provides the current needed by the loudspeaker while preserving the signal polarity and most of its voltage amplitude. These features make it suitable as the final stage of an amplifier whose preceding stages already provide the required voltage level [1, sec. 9–2].

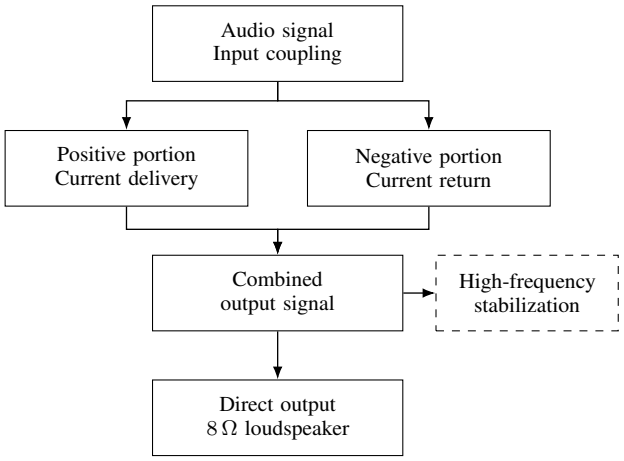


Figure 2. Functional blocks of the selected Class AB output arrangement.

III. Design Considerations and Calculated Values

A. Preamplification Stage

B. Voltage Amplification Stage

C. Class AB Power Stage

Figure 3 presents the power-stage circuit. The design retains the complementary output arrangement, with nominal supplies of $+24\,\mathrm{V}$ and $-24\,\mathrm{V}$. The load is connected directly to VOUT, the junction between RS1 and RS2.

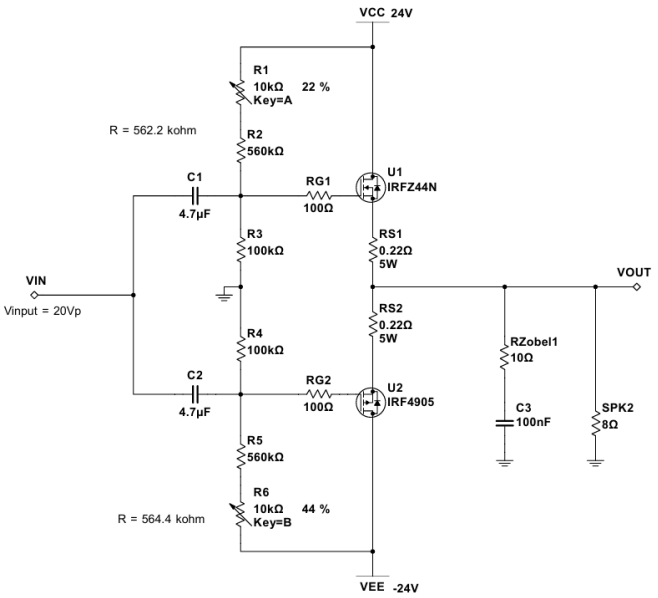


Figure 3. Class AB power-stage schematic with IRFZ44N and IRF4905 MOSFETs.

1) Output Power Requirement: The design reference is $20\,\mathrm{W}$ delivered to an $8\,\Omega$ load. For a sinusoidal signal, the average power is

$$P_L = \frac{V_{L,\mathrm{rms}}^2}{R_L}. \tag{1}$$

Here, RMS is the effective voltage value used to calculate average audio power. The required load voltage and current are

$$V_{LOAD,\mathrm{rms}} = \sqrt{(20)(8)} \approx 12.65\,\mathrm{V}, \tag{2}$$

$$V_{LOAD,peak} = \sqrt{2}(12.65) \approx 17.89\,\mathrm{V}, \tag{3}$$

$$I_{LOAD,\mathrm{rms}} = \frac{12.65}{8} \approx 1.58\,\mathrm{A}. \tag{4}$$

These values define the load requirements. With the specified $20\,\mathrm{V}$ peak input, the output power depends on the voltage transfer of the stage. Table I summarizes the calculated values.

Table I
Calculated Output Requirements for the 20 W Reference

Quantity	Calculated value
Average output power, reference	$20\,\mathrm{W}$
Load RMS voltage	$12.65\,\mathrm{V}$
Load peak voltage	$17.89\,\mathrm{V}$
Load peak-to-peak voltage	$35.78\,\mathrm{V}$
Load RMS current	$1.58\,\mathrm{A}$
Load peak current	$2.24\,\mathrm{A}$

2) Class AB Bias Adjustment: Bias sets the transistor operating conditions before an audio signal is applied. The design adopts a quiescent drain-current target of $I_{DQ} = 50\,\mathrm{mA}$ in each branch. This small resting current provides the intended overlap between the two transistors near the signal transition [1, sec. 7–2].

The $0.22\,\Omega$ source resistors provide a convenient current reference. At the selected target, the drop across each resistor is

$$V_{RS,Q} = I_{DQ}R_S = (0.050)(0.22) = 11\,\mathrm{mV}. \tag{5}$$

Bias adjustment targets these drops while keeping VOUT close to $0\,\mathrm{V}$ DC. Because SPK2 is connected directly to this node, the same DC voltage appears across the load.

The updated component labels give the effective bias resistances

$$R_N = R2 + R1_{\mathrm{eff}}, \tag{6}$$

$$R_P = R5 + R6_{\mathrm{eff}}. \tag{7}$$

The nominal gate voltages follow the two dividers:

$$V_{G,U1} = 24\,\mathrm{V}\frac{R3}{R_N + R3}, \tag{8}$$

$$V_{G,U2} = -24\,\mathrm{V}\frac{R4}{R_P + R4}. \tag{9}$$

With $R3 = R4 = 100\,\mathrm{k}\Omega$, the initial gate references of $+3.6243\,\mathrm{V}$ and $-3.6123\,\mathrm{V}$ correspond to $R_N = 562.2\,\mathrm{k}\Omega$ and $R_P = 564.4\,\mathrm{k}\Omega$. Commercial $560\,\mathrm{k}\Omega$ resistors R2 and R5 are combined with $10\,\mathrm{k}\Omega$ trimmers R1 and R6. Their nominal effective settings are $2.2\,\mathrm{k}\Omega$ and $4.4\,\mathrm{k}\Omega$, respectively, or 22% and 44% of the trimmer range.

These divider settings provide the initial gate voltages. The $50\,\mathrm{mA}$ target is assessed through the separate DC operating-point analysis and the drops in (5).

Table II
Calculated Bias References and Adjustment Targets

Quantity	Design reference
U1 gate bias, initial	$+3.6243\,\mathrm{V}$
U2 gate bias, initial	$-3.6123\,\mathrm{V}$
$R_N = R2 + R1_{\mathrm{eff}}$	$562.2\,\mathrm{k}\Omega$
$R_P = R5 + R6_{\mathrm{eff}}$	$564.4\,\mathrm{k}\Omega$
R1 effective setting	$2.2\,\mathrm{k}\Omega$
R6 effective setting	$4.4\,\mathrm{k}\Omega$
I_{DQ}, target per MOSFET	$50\,\mathrm{mA}$
RS1 and RS2 drop at target	$11\,\mathrm{mV}$ each
VOUT at rest, target	$0\,\mathrm{V}$ DC

3) Signal Coupling: C1 and C2 transfer the audio signal to the gates while keeping their DC bias settings separate from the preceding stage. Each capacitor is $4.7\,\mu$F. With a low-impedance driving source, the divider resistance seen by each capacitor is approximately $84.9\,\mathrm{k}\Omega$, giving an input coupling corner near $0.40\,$Hz. C1 and C2 are polarized electrolytics. With zero DC at VIN, the positive terminal of C1 faces the upper bias node, while the positive terminal of C2 faces VIN.

The output is connected directly to the $8\,\Omega$ load. RZobel1 and C3 form the series output RC network, with values of $10\,\Omega$ and $100\,$nF. This branch provides a high-frequency load in parallel with the loudspeaker.

4) Heat Dissipation: All thermal calculations use an ambient temperature of $T_A = 25\,^\circ$C, both at rest and during signal operation. The analysis relates each component's power loss to its estimated steady-state temperature. Table III summarizes the results and the thermal-resistance parameters used in the calculation.

Power losses: At $I_{DQ} = 50\,$mA, the resting losses in each MOSFET and source resistor are

$$P_{D,Q} = (24 - I_{DQ}R_S)I_{DQ} \simeq 1.20\,\mathrm{W}, \tag{10}$$
$$P_{RS,Q} = I_{DQ}^2 R_S = 0.55\,\mathrm{mW}. \tag{11}$$

Resistor losses follow $P_R = I_{\mathrm{rms}}^2 R$. For example, the recorded gate RMS voltage gives

$$P_{R3} = P_{R4} = \frac{14.6^2}{100000} = 2.13\,\mathrm{mW}. \tag{12}$$

The same calculation gives $1.09\,$mW in R2 and $1.08\,$mW in R5. R1 and R6 dissipate $4.28\,\mu$W and $8.50\,\mu$W, respectively.

For the gate-resistor loss calculation, the estimated RMS currents are $1.306\,$mA and $3.022\,$mA:

$$P_{RG1} = (1.306 \times 10^{-3})^2(100) \simeq 0.171\,\mathrm{mW}, \tag{13}$$
$$P_{RG2} = (3.022 \times 10^{-3})^2(100) \simeq 0.913\,\mathrm{mW}. \tag{14}$$

Capacitor losses follow $P_C = I_{C,\mathrm{rms}}^2 r_{ESR}$, where ESR represents the capacitor's series loss resistance. Using a fixed ESR of $34\,\Omega$ and estimated currents of $1.317\,$mA and $3.026\,$mA gives

$$P_{C1} = (1.317 \times 10^{-3})^2(34) \simeq 0.0590\,\mathrm{mW}, \tag{15}$$
$$P_{C2} = (3.026 \times 10^{-3})^2(34) \simeq 0.311\,\mathrm{mW}. \tag{16}$$

These gate and input-capacitor currents are calculated with a constant input-capacitance approximation for the MOSFETs.

At $10\,$kHz, the C3 loss calculation uses $\tan\delta = 0.025$:

$$X_{C3} = \frac{1}{2\pi f C3} = 159.15\,\Omega, \tag{17}$$
$$r_{ESR,3} = \tan\delta\, X_{C3} = 3.98\,\Omega. \tag{18}$$

The Zobel-branch current is

$$I_Z = \frac{13.5}{\sqrt{(10 + 3.98)^2 + 159.15^2}} \simeq 84.50\,\mathrm{mA}. \tag{19}$$

Therefore,

$$P_{\mathrm{RZobel1}} = I_Z^2(10) \simeq 0.0714\,\mathrm{W}, \tag{20}$$
$$P_{C3} = I_Z^2 r_{ESR,3} \simeq 0.0284\,\mathrm{W}. \tag{21}$$

The source-resistor sizing calculation includes the load current, the Zobel-branch current, and I_{DQ}:

$$P_{RS,\mathrm{design}} = 0.22(1.69 + 0.0845 + 0.050)^2 \simeq 0.732\,\mathrm{W}. \tag{22}$$

This power allowance applies to each $5\,$W source resistor.

Component temperatures: The passive-component temperature calculation is

$$T_i = T_A + R_{\theta i} P_i. \tag{23}$$

For example,

$$T_{RS1} = T_{RS2} \simeq 25 + (0.732)(55) = 65.3\,^\circ\mathrm{C}, \tag{24}$$
$$T_{C3} \simeq 25 + (0.0284)(1157) = 57.9\,^\circ\mathrm{C}. \tag{25}$$

The same expression is applied to the remaining passive components using the power and thermal resistance listed in Table III.

For the MOSFETs, the calculation includes a separate $4\,^\circ\mathrm{C/W}$ heat sink for each device and a $0.5\,^\circ\mathrm{C/W}$ mounting interface:

$$T_J = T_A + P_D(R_{\theta JC} + R_{\theta CS} + R_{\theta SA}). \tag{26}$$

The datasheet junction-to-case thermal resistances are $1.5\,^\circ\mathrm{C/W}$ for U1 and $0.75\,^\circ\mathrm{C/W}$ for U2 [2], [3]. Using the recorded transistor losses gives

$$T_{J,U1} = 25 + 6.58(1.5 + 0.5 + 4) \simeq 64.5\,^\circ\mathrm{C}, \tag{27}$$
$$T_{J,U2} = 25 + 6.63(0.75 + 0.5 + 4) \simeq 59.8\,^\circ\mathrm{C}. \tag{28}$$

At rest, the calculated junction temperatures are $32.2\,^\circ$C for U1 and $31.3\,^\circ$C for U2. At the $10\,$W per-device design allowance, they are $85\,^\circ$C and $77.5\,^\circ$C, below the selected $100\,^\circ$C design limit.

Table III
CALCULATED COMPONENT TEMPERATURES AT $25\,^\circ$C

Component	Power used	R_θ ($^\circ$C/W)	Temperature ($^\circ$C)
U1	6.58 W	6.00	64.5
U2	6.63 W	5.25	59.8
R1	4.28 μW	263	25.0
R2	1.09 mW	140	25.2
R3, R4	2.13 mW each	140	25.3 each
R5	1.08 mW	140	25.2
R6	8.50 μW	263	25.0
RG1	0.171 mW	140	25.0
RG2	0.913 mW	140	25.1
RS1, RS2	0.732 W each	$\simeq 55$	65.3 each
RZobel1	0.0714 W	75	30.4
C1	0.0590 mW	472	25.0
C2	0.311 mW	472	25.1
C3	0.0284 W	1157	57.9

Temperatures are steady-state analytical estimates at $25\,^\circ$C, rounded to $0.1\,^\circ$C. U1/U2 entries are junction temperatures; passive entries are body temperatures. The R_θ column lists the calculation parameters. RS1/RS2 use a power allowance for component sizing. Gate and capacitor losses are analytical estimates.

IV. Component Selection Criteria

A. Preamplification Stage

B. Voltage Amplification Stage

C. Class AB Power Stage

The selected components retain the circuit topology and use commercial nominal values. Table IV lists the main components and the ratings considered in their selection.

Table IV
Power-Stage Components and Selection Ratings

Reference	Value or model	Selection rating
U1	IRFZ44N	N-channel MOSFET
U2	IRF4905	P-channel MOSFET
R2, R5	560 kΩ	1%, 0.25 W
R3, R4	100 kΩ	1%, 0.25 W
R1, R6	10 kΩ	Trimmers, 0.25 W
RG1, RG2	100 Ω	0.25 W
RS1, RS2	0.22 Ω	5 W
C1, C2	4.7 µF	50 V, electrolytic
RZobel1	10 Ω	2 W
C3	100 nF	50 V, ceramic, code 104
SPK2	8 Ω	30 W nominal rating

The MOSFET voltage ratings are considered together with the $\pm 24\,\mathrm{V}$ supplies and the expected signal excursions [2], [3]. The source-resistor power ratings provide margin for the load current, while the gate resistors and output RC network help control high-frequency behavior.

The heat-sink selection follows the stated dissipation allowance. The component references in Table IV match the updated schematic, including C1/C2 as electrolytic coupling capacitors and C3 as the $100\,\mathrm{nF}$ ceramic Zobel capacitor marked 104.

V. Simulation Methodology

A. Common Simulation Conditions

Multisim provides the circuit environment for the reported test. The power-stage input is a $20\,\mathrm{V}$ peak sinusoidal signal at $10\,\mathrm{kHz}$, with zero DC offset. The circuit uses nominal $\pm 24\,\mathrm{V}$ supplies and an $8\,\Omega$ load. R1 and R6 have nominal settings of 22% and 44%, respectively. These settings establish the initial divider voltages; the DC analysis assesses I_{DQ} separately.

B. Preamplification Stage

C. Voltage Amplification Stage

D. Class AB Power Stage

1) DC Operating Point: The DC analysis examines the circuit with no audio input. It identifies the transistor bias and the internal output voltage. The design targets are $I_{DQ} = 50\,\mathrm{mA}$ in each branch, $11\,\mathrm{mV}$ across each source resistor, and VOUT close to zero. These references define the intended Class AB resting condition.

2) Transient Response: The transient analysis follows the input and output signals over time. It provides the voltage, current, and average-power measurements reported in the results. The method compares settled signal cycles, allowing the output level and the transition between the two transistors to be examined under the same test conditions.

3) Frequency Response and Signal Quality: Frequency-response evaluation examines how the amplifier handles signals at different audio frequencies. Waveform and harmonic analyses assess changes in signal shape, including the transition between the positive and negative portions. These assessments complement the power measurements; the numerical readings presented here correspond to the documented $10\,\mathrm{kHz}$ test.

VI. Simulated Values and Results

A. Preamplification Stage

B. Voltage Amplification Stage

C. Class AB Power Stage

Tables V and VI retain the available $10\,\mathrm{kHz}$ reference readings for comparison and thermal calculations. The updated schematic defines a single output node shared by the source-resistor junction and SPK2. RMS describes the effective signal level, peak-to-peak describes the full voltage or current swing, and mean describes the average DC component. The power readings are average values over the measurement interval.

Table V
Recorded Voltage References at 10 kHz

Location	Peak-to-peak	RMS	Mean
IRFZ44N gate	39.9 V	14.6 V	+3.62 V
IRF4905 gate	39.8 V	14.6 V	−3.62 V
Output AC reference	38.2 V	13.5 V	—

The output reference is $13.5\,\mathrm{V}$ RMS. The dash indicates that a DC measurement for the revised VOUT connection is not included in these reference readings. Its DC adjustment target is $0\,\mathrm{V}$. The gate means describe signal-on operation; I_{DQ} is assessed with the input amplitude set to zero.

Table VI
Recorded Current and Power References at 10 kHz

Quantity	Simulated value
Load-current peak-to-peak	4.78 A
Load-current RMS	1.69 A
Average loudspeaker-load power	22.8 W
IRFZ44N average dissipation	6.58 W
IRF4905 average dissipation	6.63 W
Signal frequency	10.0 kHz

The reference output-power reading exceeds the $20\,\mathrm{W}$ design reference. The transistor dissipation values describe heat produced within the output devices and are accounted for separately from the power received by the load.

VII. Analysis and Integration

A. Preamplification Stage

B. Voltage Amplification Stage

C. Class AB Power Stage

The recorded reference output voltage supports the average-power result. Applying (1) gives

$$P_L = \frac{(13.5)^2}{8} = 22.78\,\mathrm{W}, \tag{29}$$

which agrees with the $22.8\,\mathrm{W}$ simulation reading after rounding. The expected current is $13.5/8 = 1.69\,\mathrm{A}$ RMS, also consistent with the recorded value.

The $38.2\,\mathrm{V}$ peak-to-peak output corresponds to approximately $19.1\,\mathrm{V}$ peak. Compared with the specified $20\,\mathrm{V}$ peak input, the output amplitude is 95.5% of the input amplitude. This small reduction is consistent with the power stage's role of supplying current while preserving the signal polarity and most of its voltage amplitude.

In the recorded reference test, the delivered power is 14% above the $20\,\mathrm{W}$ objective. This result provides the comparison baseline for the updated power-stage schematic. The two transistor losses are similar, and both remain below the $10\,\mathrm{W}$ per-device thermal design allowance.

For integration with the complete amplifier, the preceding stage provides the required input amplitude while driving the power-stage input. The bias references describe the resting condition, while the reported voltage, current, and power values describe operation with the audio test signal applied.

VIII. Conclusions and Next Steps

A. Preamplification Stage

B. Voltage Amplification Stage

C. Class AB Power Stage

The complementary IRFZ44N/IRF4905 stage retains the Class AB topology and uses commercial component values. Its bias design includes a $I_{DQ} = 50\,\mathrm{mA}$ resting-current target, with trimmers for adjustment and source resistors that provide current references. C1 and C2 couple the input signal to the transistor gates, while the load connects directly to VOUT. The thermal assessment covers transistor, resistor, and capacitor losses and gives calculated temperature estimates for the stage electronics.

With a $20\,\mathrm{V}$ peak input at $10\,\mathrm{kHz}$, the recorded reference test delivered $22.8\,\mathrm{W}$ to an $8\,\Omega$ load, exceeding the $20\,\mathrm{W}$ design reference. The output reaches $13.5\,\mathrm{V}$ RMS and $1.69\,\mathrm{A}$ RMS. These reference results support the power-delivery objective for the documented test. The updated DC assessment uses the direct output connection and the $50\,\mathrm{mA}$ adjustment target. The thermal calculation gives MOSFET junction temperatures of $64.5\,^{\circ}\mathrm{C}$ and $59.8\,^{\circ}\mathrm{C}$. Frequency-response, signal-quality, and connected-stage assessments complete the evaluation framework for the amplifier.

References

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[3] Infineon Technologies, "IRF4905PbF: HEXFET power MOSFET," datasheet, rev. 2.1, Apr. 8, 2026, pp. 1–4.